Chapter 11. In the browser

This chapter covers

- Building browser-based programs
- Creating polyfills for cross-browser compatibility
- Building retained- and immediate-mode interfaces
- Structuring browser-based programs

A web browser can be a hostile environment in which to run a program, but CoffeeScript can help you keep your browser-based applications manageable. It's not the individual features of CoffeeScript that help you in a browser. Instead, it's the focus on simplicity and clarity of expression and semantics that help you to keep your sanity in the world of web browsers, which can at times seem insane. That said, although the benefit of CoffeeScript in a browser is in how you use it to structure programs, it's still important to learn the challenges of the browser environment that your CoffeeScript programs live in.

In this chapter you'll learn how to write browser-based programs in CoffeeScript that run on multiple, often incompatible, browsers. To get there you'll learn how to deliver your CoffeeScript to a browser, how to deal with browsers that don't support the features you need, how to create user interfaces using retained- and immediate-mode techniques, how to create animations, and how to manage the structure of your program and how fast it runs. It's time to get started.

11.1. Getting started

Imagine Agtron has a few servers that frequently run out of both CPU and network bandwidth. This causes his online shop to temporarily go offline, and Agtron loses money. If only Agtron had a web-based dashboard to show him the status of his servers (in real time), then he could prevent the shop from being offline.